

Ivan Grigorik

Ph.D. Student at UT Austin

+1 (267) 895 4294
grigorik@utexas.edu
ivangrigorik.github.io
in grigorik
IvanGrigorik
IvanGrigorik

Research Interests

High Performance Computing, Compilers, Debuggers, Software Testing and Verification, and AI for SE

Education

2024–present **Ph.D.**, *The University of Texas at Austin*, Austin, United States, *GPA: 3.9*
Major in Software Engineering and Systems, advised by Milos Gligoric.

2020–2024 **B.S.**, *Belarusian State University of Informatics*, Minsk, Belarus, *GPA: 3.7*
Major in Computer Science and Network Engineering

Skills

Languages C, C++, Python, Rust, ASM, Java, Bash, SQL;
Tools GCC, LLVM, Make & CMake, CUDA, GDB, Valgrind, Docker, STL,
Domains Compilers, HPC, Static and Dynamic Analysis, Verification of Software;
Concepts System and Software Engineering, Program Testing, Sanitizing and Profiling, Design Patterns, Distributed Computing, Parallel Algorithms, Multithreading, Computer Architecture, Networking.

Experience

Summer 2024 **Research Assistant**, *University of Texas at Austin*, Austin

- present Working on Software Testing and Validation and Compilers tools

- Researching compiler optimization techniques, source code processing, and code testing tools;
- Developing static and dynamic analyzer tools for the existing programming language on top of its infrastructure;
- Researching the source-to-source code conversions in HPC languages;
- Experimenting with prompt engineering and LLM-based tooling for research;
- Working with HPC code conversions and optimization techniques (PyKokkos, Kokkos);
- Working on HPC eDSL debuggers.

Spring - **Research Assistant**, *BSUIR*, Minsk, Belarus

Summer 2024 Led the research on wireless communication technology

- Conducted in-depth research on the topic of harmonic oscillations and total harmonic distortion between signals in wireless networks;
- Designed and developed the PCB for jamming Bluetooth and WiFi devices;
- Contributed to 3 different research projects over 5 months.

Summer 2023 **Software Developer Intern**, *AGAT - Control Systems*, Minsk, Belarus

Contributed to multiple network tools for the company network.

- Developed network controlling and monitoring software for Cisco and Dell equipment.
- Implemented and designed network analysis tools for the company's local network.
- Collaborated with the team to deploy an improved network management system.

Projects

xbash A fork of bash with static and dynamic analysis tools and extended builtins.
cuRTX Open-source ray tracing tool written in CUDA and C++, inspired by RTX-in-one-weekend book
GameOfLife Pet-project, Conway's Game of Life written in C++ + SFML
HvC Fork of HIPIFY that compares heterogeneous computing platforms (CUDA, HIP, etc.)
PyKokkos Framework for writing high-performance Python code similar to Numba. *Current maintainer

Courseworks

- Programming languages and computer design
- Computer Architecture
- Software Networks
- Compilers
- System software
- HPC architecture
- Hardware Systems
- Programming Paradigms

Honors and awards

- 2024 Graduate School Fellowship, University of Texas at Austin
- 2024 Outstanding paper of BSUIR 60th SE
Paper: Active jammer stations based on direct influence noises

Services and event participation

- 2025-2026 CGO '25 sub-reviewer, OOPSLA '26 sub-reviewer

Presentations

- 2025 AMD iMAGiNE Symposium.
- 2026 Predictive Science Academic Alliance Program (PSAAP) presentation: PyKokkos ecosystem and development cycle.
- 2026 AMD iMAGiNE Symposium participation
- 2026 High Performance Software Foundation (HPSF) presentation: Pythonizing PyKokkos.
- 2024-present Co-organized the Joint UT-Cornell Software Engineering Seminar Series